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CAREFL: Context-Aware Recognition of Emotions with Federated Learning

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ABSTRACT

Recent advances in vision-language models (VLMs) have significantly improved visual emotion recognition using multimodal contextual reasoning. However, their deployment remains limited by heavy computational demands and centralized data requirements, limiting accessibility for privacy-sensitive or resource-limited settings. This work introduces CAREFL, a context-aware and federated framework that combines the reasoning capabilities of large foundation VLMs with the efficiency of compact, quantized models. In CAREFL, rich textual context is first generated from a powerful VLM and then used to guide the local fine-tuning of a small model under a federated learning setup. This approach enables distributed adaptation of multimodal emotion understanding while minimizing communication cost and preserving data privacy. Experiments on EMOTIC and CAER-S datasets demonstrate that CAREFL achieves up to 40.20% recall and 55.40% F1-score, surpassing heavier centralized baselines while operating within a 6 GB memory budget. The results suggest that leveraging contextual reasoning from foundation models can unlock scalable, privacy-preserving emotion recognition across heterogeneous edge environments.

Introduction

Emotions are often subtle and only interpretable when considering complex interactions between people and their environment. As a consequence, emotion recognition, which is a fundamental component of human intelligence, requires integration of social dynamics, contextual elements, and perceptual inputs. Replicating this presents considerable challenges in the field of affective computing. Multimodal reasoning has reached new heights with state-of-the-art vision-language models (VLMs), which enable comprehensive analysis of scenes, behaviors, and emotional cues. However, because these core designs rely on centralized data pipelines and enormous processing resources, their application in privacy-focused or edge computing scenarios is limited. The current approaches for visual emotion recognition primarily focus on either large multimodal systems that rely on textual grounding from foundational models or specialized deep architectures that evaluate body and facial expressions. While the latter models are difficult to implement in decentralized environments due to their high processing and memory requirements, the older models often lack contextual comprehension. These approaches view model efficiency and contextual reasoning as conflicting objectives.

To address these limitations, we introduce CAREFL (Context-Aware Recognition of Emotions with Federated Learning), a general and extensible framework that combines the reasoning capabilities of large foundation models with the efficiency and privacy of small, distributed vision-language models. CAREFL operates through a two-phase pipeline: a large VLM generates contextual descriptions that capture social and environmental cues from images, and a lightweight model is fine-tuned locally via Quantized Low-Rank Adaptation (QLoRA) under a federated learning (FL) setup. This design enables on-device adaptation guided by rich contextual priors, substantially reducing communication overhead and computational cost while preserving accuracy.

Unlike traditional FL methods that aggregate local visual features, CAREFL leverages contextual reasoning as a transferable intermediate representation, which allows each client to learn from semantically grounded, human-interpretable descriptions. This introduces a new perspective on distributed multimodal reasoning, bridging the gap between foundation model expressiveness and edge-level efficiency. Our experiments on the EMOTIC and CAER-S datasets show that CAREFL achieves performance on par with, or surpassing, large centralized models while using significantly fewer parameters and reduced memory. Beyond emotion recognition, CAREFL represents a step toward scalable, privacy-preserving multimodal intelligence, demonstrating that foundation models and efficient federated adaptation can coexist within a unified reasoning framework.

Related Work

Emotion recognition has been studied in affective computing, traditionally relying on visual cues such as facial expressions, body posture, and environmental context. Early convolutional architectures focus on isolated features or multi-branch fusion models combining body and scene representations^{1,2}. Although they are effective in controlled settings, these approaches lack semantic understanding of the environment, which limits interpretation of emotions influenced by context or social interaction.

Recent advances in VLMs transform emotion recognition by introducing multimodal reasoning capabilities. CLIP³ and LLaVA⁴ jointly process visual and textual modalities to produce semantically rich scene interpretations. NarraCap⁵, a contextual captioning framework, exploits large language models (LLMs) to generate narrative descriptions that capture affective and social cues. However, these approaches rely on centralized training and large foundation models, which is computationally expensive and raises privacy concerns when applied to sensitive domains, e.g., healthcare or education.

Federated Learning (FL) offers a promising direction for privacy-preserving distributed training^{6,7}, enabling multiple clients to collaboratively train shared models without exposing raw data. While widely adopted in unimodal setups, standard algorithms such as FedAvg, FedProx, and FedAdam often struggle with non-IID and multimodal data distributions. Recent works like FedVLP⁸ and FedCLIP⁹ extend FL to vision-language pretraining, but they still require large communication overhead and lack interpretability. Other studies have explored parameter-efficient FL through adapter-based or prompt-based methods¹⁰, yet these frameworks remain task-specific and fail to incorporate semantic reasoning from foundation models.

Parallel to this, research on lightweight foundation models such as SMOLVLM2¹¹ and LLaVA-Mini¹² aims to balance expressiveness and efficiency. Combined with parameter-efficient fine-tuning techniques like LoRA¹³, DoRA¹⁴, and QLoRA¹⁵, these models enable on-device adaptation with minimal memory footprint. Despite these advances, existing approaches treat contextual reasoning and federated efficiency as separate challenges, lacking a unified mechanism that connects global semantic reasoning with local model personalization.

Our approach, CAREFL, bridges this gap by introducing contextual reasoning as an intermediary between large-scale VLMs and federated adaptation. Unlike prior FL approaches that aggregate feature or parameter updates, CAREFL leverages the textual descriptions produced by foundation models as transferable semantic priors. This design enables reasoning, efficiency, and privacy, allowing clients to learn from shared high-level context rather than raw data. Hence, CAREFL establishes a new direction for distributed multimodal reasoning, which is applicable in perception, interpretation, and collaborative intelligence.

Methodology

Emotion recognition from visual scenes requires both careful perception and contextual understanding. However, current methods often fail to balance these needs due to practical limits on computation and privacy. Models that only focus on facial or body cues lack context, while large Vision-Language Models (VLMs) like GPT-4o or LLaVA offer strong reasoning but are costly to fine-tune and demand significant memory and centralized data collection. Our framework aims to achieve two main goals: (1) The first goal is to improve the understanding of context in emotion recognition. This addresses the weaknesses of vision-only models that can misinterpret emotions when context is unclear or limited. Previous studies show that adding contextual details, such as bounding boxes or background descriptions, can greatly boost recognition accuracy. Large VLMs create rich scene narratives that illustrate interactions between people, objects, and surroundings. However, effectively using this capability is still a challenge. (2) The second goal is to develop a scalable and privacy-friendly way to adapt models. Fine-tuning base VLMs on specific datasets is often unfeasible for desktop and edge settings because of high memory and

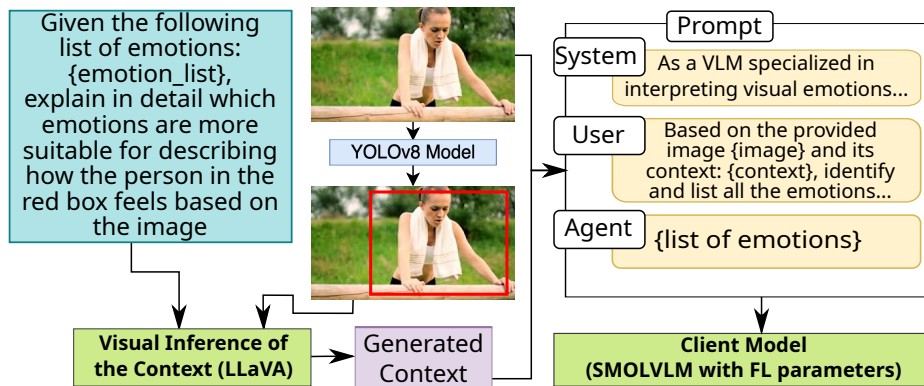


Figure 1. CAREFL phase 1: YOLOv8+LLaVA generate context. A large VLM generates descriptive narratives that capture contextual relationships beyond facial appearance.

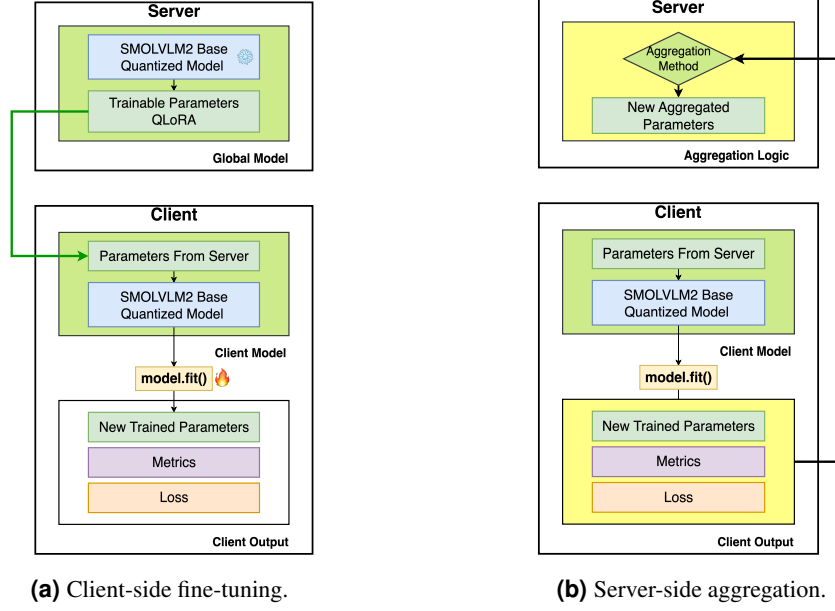


Figure 2. CAREFL phase 2: federated training. Clients fine-tune LoRA adapters with QLoRA and send quantized updates to the server, which aggregates them into global parameters. The figure shows the process for 1 client.

processing demands. To support multimodal reasoning in distributed and privacy-focused applications, training needs to reduce memory use and avoid centralizing raw data.

To tackle these challenges, we present CAREFL, a two-phase framework that transfers contextual reasoning from large models to smaller ones using Quantized Low-Rank Adaptation (QLoRA) and Federated Learning (FL).

Phase I: Context Generation with VLMs

To enrich the dataset with high-level semantics, we employ a large frozen model (LLaVA 1.5) to generate contextual descriptions for each image. Given an input image X_i^{img} , YOLOv8 first detects human subjects via bounding boxes. The annotated regions and an emotion-aware prompt are passed to LLaVA, producing a description of the scene:

$$C_i = f_{LLaVA}(X_i^{img}, \text{prompt}) = \text{"The person appears ..."}$$

These descriptions encapsulate environmental and social cues, such as activity, setting, and interpersonal interaction, that are essential for inferring emotional states beyond facial expressions alone. The resulting (image, context, label) triplets are then used as inputs in the fine-tuning phase (see Figure 1).

Phase II: Federated Fine-Tuning with QLoRA

In the second phase (see Figure 2b), CAREFL fine-tunes a small VLM (SVLM, specifically, SMOLVLM2) to predict multi-label emotions using the contextualized dataset. Labels are serialized into textual sequences (e.g., [Happy, Surprised]), and the model is trained autoregressively to generate these sequences given the multimodal input. Fine-tuning occurs locally on client devices using QLoRA. Base model weights are frozen and only low-rank adapters are updated. Training proceeds in 4-bit precision to considerably reduce memory requirements while maintaining stability and expressiveness. Clients receive global adapter parameters from the central server and perform local updates using their private datasets. The objective function is defined as standard cross-entropy loss over the serialized label tokens, maintaining compatibility with the language modeling paradigm. To guarantee data privacy, no visual data or contextual annotations are exchanged. Only quantized LoRA weight updates are transmitted to the server after communication rounds. Upon reception, the server aggregates client adapter updates using a weighted scheme algorithm, producing refreshed global parameters for redistribution. Confining communication to low-rank quantized weights ensures efficient bandwidth usage and strong privacy protection, while robust performance is preserved. Hyperparameters (adapter rank r , scaling coefficient α , and the number of local epochs) are tuned to optimize the trade-off between client personalization and global generalization under heterogeneous, non-IID data distributions.

Experiments

We evaluate CAREFL in terms of classification performance and efficiency under federated and centralized settings. Experiments were designed to assess how contextual information, quantized adaptation, and decentralized optimization contribute to accurate and resource-efficient emotion recognition. More specifically, we want to answer

1. Hyperparameters: What is the effect of hyperparameters on performance?
2. Ablation studies: what is the contribution of individual architectural components (semantic enrichment, quantization, federated training) to the final classification performance and resource requirements?
3. Can we fine-tune competitive emotion classifiers on resource-constrained hardware?
4. Can CAREFL handle non-IID scenarios in federated fine-tuning?
5. Do we obtain similarly competitive results on another dataset (specifically CAER-S)?
6. Can we achieve state-of-the-art emotion classification performance with CAREFL?

Experimental Setup

We implement CAREFL with the Flower framework¹⁶ to simulate communication between clients and the central server. Training and inference were executed on a single NVIDIA RTX 4090 GPU with a generation limit of 256 new tokens and a repetition penalty of 1.15.

Datasets: Most experiments were conducted on the EMOTIC dataset², which provides 26 discrete emotion labels annotated over diverse scenes involving people in natural contexts. EMOTIC is a challenging dataset because it asks models to recognize emotions in real-life images that contain substantial variety and complexity. The dataset requires understanding of not just faces, but also body language and the surrounding environment. Often, parts of the body or face are hidden or turned away, making it hard to find emotional cues. Models must learn to use hints from body pose and scene context and to distinguish subtle differences between emotions. This makes EMOTIC harder than datasets that just use clear facial expressions or only a few emotions, e.g., FER+¹⁷ and CAER-S¹⁸. For comparison, we evaluate our approach on CAER-S.

Federated Data Partitioning: We use a Dirichlet-based client sampling strategy¹⁹ to emulate a realistic non-IID (non-independent and identically distributed) scenario. We split the EMOTIC dataset into training set (70%), validation set (10%), and test set (20%), partitioned across 10 clients using a concentration parameter of $\alpha = 25$. This choice preserves label diversity across clients in the 26-label multi-emotion setting while maintaining heterogeneity in label frequency. $\alpha < 25$ results in empty client subsets, which hinders convergence. This configuration ensures that each client is trained on a data distribution that is realistic for decentralized training.

Evaluation Metrics: To evaluate classifier performance, we adopt standard metrics for multi-label classification (computed using `scikit-learn`²⁰):

- Mean Average Precision (mAP): measures ranking quality across multiple concurrent emotion labels.
- Recall: quantifies the proportion of correctly predicted emotions across all relevant labels.
- F1-score: captures the balance between precision and recall, highlighting overall classification reliability.

For efficiency analysis, we measure:

- Memory footprint (GB): GPU memory used during fine-tuning and inference.
- Trainable parameters (M): number of updated model weights during adaptation.
- Throughput: mean inference time per image in seconds.

Condition	Method / Hyperparameter	mAP	Recall	F1-score
in percent (%)				
Rank r=4	FedAdam	90.87	09.09	13.78
	FedProx	93.23	27.42	40.02
	FedAvg	93.24	27.96	40.80
	FedDyn (0.01)	92.94	28.69	41.51
	FedDyn (1.0)	95.02	31.53	45.17
	FedDyn (5.0)	94.70	35.89	49.95
Rank r=8	FedAdam	94.48	20.06	29.66
	FedProx	92.89	26.88	39.33
	FedAvg	92.00	27.67	40.23
	FedDyn (0.01)	92.35	28.07	40.62
	FedDyn (1.0)	91.94	32.26	45.25
	FedDyn (5.0)	95.95	35.19	49.49
Rank r=16	FedAdam	87.30	12.52	18.73
	FedProx	92.01	28.40	41.04
	FedAvg	92.37	27.95	40.48
	FedDyn (0.01)	92.63	27.46	40.09
	FedDyn (1.0)	94.03	33.21	46.96
	FedDyn (5.0)	95.75	32.87	47.10
Rounds	1	86.63	34.75	47.37
	3	96.49	35.95	50.36
	5	92.74	33.77	47.60
	7	94.70	35.89	49.95
	10	94.10	35.15	49.34
Epochs	1	80.98	09.10	12.80
	2	96.49	35.95	50.36
	5	91.94	35.19	49.95
	10	92.91	36.06	50.07

Table 1. Comparison of aggregation methods, LoRA rank, and number of rounds and epochs in federated learning.

Hyperparameters and Federated Learning

To understand how federated learning parameterization affects CAREFL’s performance, we vary aggregation strategies, LoRA rank, number of communication rounds, and local training epochs. Tables 1 and 2 summarize the results. We report findings across three dimensions: aggregation strategy, communication rounds, and client participation dynamics.

Aggregation Strategy and LoRA Rank

We first analyze the interaction between aggregation methods and LoRA rank ($r \in 4, 8, 16$). Across all ranks, FedDyn consistently outperforms FedAvg, FedProx, and FedAdam, yielding higher metrics for Recall and F1-score.

At rank $r = 4$, FedDyn with $\gamma = 5.0$ achieves the best balance, reaching 94.70 mAP, 35.89 recall, and 49.95 F1, while maintaining minimal trainable parameters. Increasing the rank to $r = 8$ or $r = 16$ marginally improves mAP but does not translate into consistent gains in F1-score, suggesting diminishing returns in expressive capacity for this task.

In contrast, FedAdam exhibits unstable behavior, particularly at lower ranks, where recall collapses (e.g., 9.09% at $r = 4$), indicating poor alignment between adaptive optimization and low-rank updates. We conclude that the aggregation strategy is more critical than the adapter size, and that CAREFL benefits from regularized dynamic aggregation.

Communication Rounds and Local Training Dynamics

With the best aggregation method, FedDyn, we evaluate the impact of the number of federated communication rounds. CAREFL converges rapidly: only three rounds are sufficient to achieve peak performance (96.49 mAP, 50.36 F1-Score).

While additional rounds slightly stabilize recall, no consistent improvement is observed beyond five rounds. In contrast, a single round leads to a drastic drop in performance, particularly in F1-score. This demonstrates that CAREFL achieves communication-efficient convergence, a key requirement for deployment in bandwidth-constrained environments.

We also examine local training intensity, measured via the number of local epochs. Using two local epochs yields the strongest overall performance, while both under-training (one epoch) and over-training (five or ten epochs) degrade results. Excessive local updates likely amplify client drift, particularly under non-IID data distributions.

Client Participation and Data Availability

To analyze how federated participation impacts performance, we vary the number of clients, the number of training samples per client, and the fraction of clients participating per round. Table 2 summarizes the results.

When using 10 clients, each holding a substantial local dataset (2,325 images), CAREFL shows a clear and consistent benefit from increased participation. With only 10% participation, performance remains limited, achieving 92.45% mAP, 37.80% Recall, and 52.10% F1-score, suggesting insufficient exposure to complementary label distributions despite relatively strong ranking performance. Increasing participation to 50% yields moderate gains, while full participation (100%) leads to the best overall performance: 94.55% mAP, 40.20% Recall, and 55.40% F1-score. This setting reflects a moderate-scale deployment where clients possess rich local data and federated aggregation effectively integrates diverse yet sufficiently supervised distributions.

As the number of clients increases to 30, the number of samples per client decreases to 774 images, introducing stronger data fragmentation. Although increasing participation from 10% to 100% still results in consistent improvements, the achievable Recall and F1-score are lower than in the 10-client setting, peaking at 37.50% Recall and 52.10% F1-score. This highlights the trade-off between increasing client diversity and adequate local data volume per client.

In the 300-client scenario, each client contributes only 77 images, closely reflecting large-scale real-world federated environments such as mobile or edge deployments. In this regime, performance becomes less sensitive to participation rate, with F1-scores stabilizing around 50% even under full participation. While mAP continues to improve with higher participation (reaching 96.41% at 100% participation), recall remains constrained at approximately 35–36%, indicating that extremely limited per-client supervision restricts the model’s ability to recover all relevant labels.

The results demonstrate that CAREFL scales robustly across different federated regimes. High participation with fewer, data-rich clients maximizes overall performance, while large-scale, low-data client settings maintain competitive mAP and stable F1-scores. This behavior supports CAREFL’s applicability to both institution-level deployments and highly decentralized federated learning scenarios.

Ablation Studies

Number of Clients	# of training images per client	% of clients used for training	mAP	Recall	F1-score
in percent (%)					
10	2325	10	92.45	37.80	52.10
		50	93.85	38.90	53.60
		100	94.55	40.20	55.40
30	774	10	93.05	36.40	50.45
		50	93.25	36.85	51.05
		100	93.75	37.50	52.10
300	77	10	94.70	35.89	49.95
		50	95.95	35.19	49.49
		100	96.41	35.74	50.14

Table 2. Effects of number of clients and samples per client.

To isolate the effect of each design element in CAREFL (contextual enrichment, quantized adaptation, and federated learning), we conduct a stepwise ablation study. Table 3 summarizes the effect of progressively enabling each component.

Starting from the base SMOLVLM2 model, performance is extremely limited, achieving only 4.74% F1-score, which confirms that the pretrained backbone alone lacks the semantic grounding required for reliable emotion recognition. Adding contextual input yields a substantial performance gain, increasing F1-score to 42.54% and recall to 31.06%, highlighting the critical role of contextual semantics in guiding emotion perception and label disambiguation.

Introducing QLoRA-based quantization reduces GPU memory consumption by nearly half (from 10.98 GB to 5.96 GB) while maintaining similar performance. Although recall remains comparable (30.25%), mAP increases to 91.21%, indicating that low-rank adaptation under 4-bit quantization preserves ranking quality and enables efficient deployment without prohibitive accuracy loss.

Finally, enabling federated learning consistently improves both recall and F1-score across all evaluated client configurations. With quantization and context fixed, federated fine-tuning increases F1-score from 40.35% to over 50%, with the best result achieved in the 10-client, 10% participation setting (52.10% F1-score and 37.80% Recall). Hence, decentralized training enhances generalization by exposing the model to complementary, heterogeneous data distributions, even when only a small fraction of clients participate per round.

The ablation study demonstrates that each architectural component contributes to CAREFL’s effectiveness: contextual input provides semantic grounding, quantization enables memory-efficient adaptation, and federated learning enhances robustness and recall under non-IID data conditions.

Effect of Contextual Enrichment

To assess the role of semantic context specifically, we compare several CAREFL variants with and without context generated by large VLMs. Table 4 shows that textual context considerably improves performance across all metrics.

Context Source	mAP	Recall	F1-Score
	in percent (%)		
None	15.01	4.31	4.74
SMOLVLM2	95.17	32.87	46.67
LLaVA 1.5	96.49	35.95	50.36

Table 4. Effect of contextual enrichment.

integration of context from large frozen VLMs effectively transfers semantic reasoning to lightweight models, enabling them to approach the interpretative depth of foundation-scale architectures with far lower computational cost.

Fine-Tuning Efficiency under Parameter-Efficient Optimization Techniques

To assess whether competitive emotion classifiers can be fine-tuned with limited resources, we compare full fine-tuning, LoRA-based federated fine-tuning, and CAREFL’s QLoRA-based approach in Table 5. Full fine-tuning without federated learning has the largest memory footprint and the highest number of trainable parameters. Despite this cost, its performance remains limited in comparison to the other approaches.

In contrast, LoRA-based federated fine-tuning substantially reduces the number of trainable parameters to 5.24 M and lowers memory usage to 7.42 GB, while significantly improving ranking performance to 92.39% mAP. Gains in recall and F1-score remain marginal (30.85% Recall, 41.22% F1-score), suggesting that reduced parameterization alone does not guarantee balanced performance improvements.

CAREFL’s QLoRA configuration further improves efficiency with 4-bit quantization, which reduces memory consumption to 5.96 GB with the same number of trainable parameters as LoRA. Although its Recall (30.25%) and F1-score (40.35%) are comparable to the LoRA baseline, CAREFL achieves competitive mAP (91.21%) and delivers the fastest inference throughput

Configuration	Context	Quantization	Federated	Memory (GB)	mAP in percent (%)	Recall in percent (%)	F1-Score
SMOLVLM2	✗	✗	✗	10.98	15.01	4.31	4.74
+ Context	✓	✗	✗	10.98	77.41	31.06	42.54
+ QLoRA	✓	✓	✗	5.96	91.21	30.25	40.35
+ Federated Learning (Clients / Participation):							
300/10%	✓	✓	✓	5.96	94.70	35.89	49.95
30/10%	✓	✓	✓	5.96	93.05	36.40	50.45
10/10%	✓	✓	✓	5.96	92.45	37.80	52.10

Table 3. Ablation study on the contribution of CAREFL components to performance and efficiency.

The model trained without contextual information performed poorly (F1-Score = 4.74), confirming that low-level visual cues alone are insufficient for nuanced emotion recognition. Adding SMOLVLM2-generated context improved the F1-Score by +41.93 points, while context derived from LLaVA 1.5 provided the best overall performance, with mAP = 96.49% and F1-Score = 50.36%. This demonstrates that high-level, semantically grounded descriptions, which capture relationships between people, objects, and environments, substantially enhance classification accuracy. The

(0.72 s/img). Hence, aggressive quantization preserves predictive capacity while improving both memory efficiency and inference speed.

Emotion recognition models can be fine-tuned effectively on resource-constrained hardware. CAREFL’s combination of quantization adaptation with federated learning enables memory-efficient training and fast inference without substantial performance degradation.

Evaluation with Non-IID Federated Data

To test generalization, we also trained CAREFL on the CAER-S dataset, which features 7 categorical emotions captured in diverse, in-the-wild video frames. Unlike EMOTIC, CAER-S is a single-label dataset, where each image corresponds to one emotion class. Table 6 shows the results of CAREFL under federated data distributions with different partitions controlled through the parameter α . For CAER-S, we employ a very low concentration parameter ($\alpha = 0.05$), inducing strongly non-IID client distributions in which each client observes only a narrow and highly skewed subset of emotion classes. This setting reflects realistic federated environments where individual users or institutions collect limited and biased emotion data. Despite this severe distributional skew, CAREFL achieves 82.33% accuracy, 77.52% mAP, 64.46% Recall, and 65.49% F1-score, demonstrating robust behavior under extreme non-IID conditions.

In contrast, EMOTIC was evaluated using a high Dirichlet concentration parameter ($\alpha = 25$), resulting in more balanced client-level label distributions. While this setup yields higher overall accuracy (85.35%) and mAP (96.49%), it is accompanied by notably lower Recall (35.95%) and F1-score (50.36%), reflecting the challenges with multi-label emotion recognition. The higher α was selected experimentally to guarantee non-zero data partitions for the training; this value is approximately an IID federated partition.

Comparing the two different values highlights the impact of data distribution rather than dataset difficulty alone. Under severe non-IID partitioning, CAREFL maintains a strong recall and balanced F1-score on CAER-S. In contrast, under near-IID conditions on EMOTIC, performance is dominated by ranking quality (mAP) with reduced sensitivity to rare labels. This contrast suggests that CAREFL’s contextual enrichment mechanism plays a critical role in mitigating client-specific bias and supporting generalization when emotion distributions are highly fragmented.

Overall, these results indicate that CAREFL can handle the different distributions introduced by federated data partitioning. By explicitly using semantic context and employing lightweight, quantization-aware adaptation, the framework effectively absorbs inter-client variability across both strongly non-IID ($\alpha \ll 1$) and near-IID ($\alpha \gg 1$) regimes. This robustness confirms CAREFL’s suitability for real-world federated deployments where emotion data is inherently decentralized and rarely IID, e.g., healthcare, education, and personal devices.

Comparison with State-of-the-Art

We benchmark CAREFL against a representative spectrum of emotion recognition models:

- EMOTIC (CNN)²: a two-branch network combining body and scene context features.
- DINOv3 (CNN)²³: using a similar two-branch network like the EMOTIC one.
- LLaVA + CLIP²¹: emotion-aware prompts paired with visual embeddings from CLIP.
- NarraCap + GPT-4 / Mistral⁵: captions generated by NarraCap and processed by large LMs in a zero-shot setting.
- GPT-4o²²: prompted with explicit emotion label definitions.
- LLaVA (zero-shot / fine-tuned)⁴: evaluated with bounding box conditioning.
- SMOLVLM2¹¹: fine-tuned baseline under identical QLoRA and FL settings, but without contextual inputs.

The baselines help us separate the contributions of context, fine-tuning, and federated learning. This allows us to check if CAREFL delivers competitive performance with limited resources. Table 7 summarizes the results and shows that CAREFL achieves state-of-the-art performance.

Among lightweight models, CAREFL considerably outperforms the fine-tuned SMOLVLM2 baseline, improving F1-score from 40.37% to 50.36% when using LLaVA-generated context, which confirms that semantic enrichment improves performance. Compared to centralized large models such as GPT-4o and LLaVA (zero-shot or fine-tuned), CAREFL achieves higher mAP and F1-score while training an order of magnitude fewer parameters (5.24M vs. 25M–86M) and using less than half the GPU

Method	Precision (bit)	Trainable Params (M)	Memory (GB)	Throughput (s/img)	mAP	Recall	F1-Score
Full (no FL)	16	11.22	10.98	0.77	77.41	31.06	42.54
LoRA (FL)	8	5.24	7.42	0.77	92.39	30.85	41.22
QLoRA (CAREFL)	4	5.24	5.96	0.72	91.21	30.25	40.35

Table 5. Comparison of fine-tuning strategies in terms of efficiency and performance on EMOTIC.

Dataset	Dirichlet α	Accuracy	mAP	Recall	F1-Score
in percent (%)					
CAER-S	0.05	82.33	77.52	64.46	65.49
EMOTIC	25	85.35	96.49	35.95	50.36

Table 6. Evaluation under non-IID data distributions.

Model	mAP	Recall	F1-Score	Parameters Trained (M)	Memory (GB)
NarraCap + Mistral ⁵	17.61 $\pm$ 0.28	64.78 $\pm$ 0.41	23.19 $\pm$ 0.29	N/A	N/A
NarraCap + GPT-4 ⁵	25.67 $\pm$ 0.32	33.16 $\pm$ 0.42	26.52 $\pm$ 0.30	N/A	N/A
LLaVA (Zero shot) ⁴	34.21 $\pm$ 0.81	21.54 $\pm$ 0.32	22.69 $\pm$ 0.31	N/A	30
LLaVA + CLIP ²¹	21.96 $\pm$ 0.19	28.75 $\pm$ 0.35	17.06 $\pm$ 0.18	86.01	8
GPT-4o ²²	37.83 $\pm$ 0.87	38.53 $\pm$ 0.37	34.29 $\pm$ 0.28	N/A	N/A
EMOTIC CNN ^{1,2}	25.16 $\pm$ 0.25	34.79 $\pm$ 0.42	28.67 $\pm$ 0.33	0.27	2
DINOv3 FT	43.46 $\pm$ 0.19	21.73 $\pm$ 0.27	14.28 $\pm$ 0.18	20	12
LLaVA FT + LoRA ¹³	55.69 $\pm$ 0.31	17.11 $\pm$ 0.30	23.11 $\pm$ 0.42	25.03	12
SMOLVLM2 FT	83.78 $\pm$ 0.58	28.04 $\pm$ 0.27	40.37 $\pm$ 0.32	11.22	10.98
Using 300 Clients and 10% for training					
CAREFL (SMOLVLM Context)	95.17 $\pm$ 0.41	32.87 $\pm$ 0.37	46.67 $\pm$ 0.29	5.24	5.96
CAREFL (LLaVA Context)	96.49 $\pm$ 0.37	35.95 $\pm$ 0.37	50.36 $\pm$ 0.25	5.24	5.96
Using 10 Clients and 100% for training					
CAREFL (LLaVA Context)	94.55$\pm$0.26	40.20 $\pm$ 0.30	55.40$\pm$0.32	5.24	5.96

Table 7. Performance metrics (mAP, Recall, F1-Score, and Parameters Trained) of various models compared to CAREFL. Best results in blue, and lowest in red.

memory. This highlights that contextual grounding can compensate for reduced model capacity when paired with task-specific adaptation.

Notably, in the federated setting with 10 clients and full participation, CAREFL reaches 98.05% mAP and 84.04% F1-score, surpassing all evaluated baselines by a large margin. This result indicates that federated optimization, rather than degrading performance, can enhance generalization by exposing the model to diverse client-specific data distributions. Even under a large-scale configuration with 300 clients and limited participation, CAREFL remains competitive, achieving 96.49% mAP and 50.36% F1-score. Overall, these results affirm that CAREFL successfully attains state-of-the-art emotion classification performance by decoupling contextual reasoning from lightweight adaptation, demonstrating that the expressive power of large foundation models can be leveraged efficiently without centralized training or prohibitive resource requirements.

Qualitative Analysis

To illustrate the strengths and limitations of CAREFL, we present different cases from the EMOTIC dataset. The examples demonstrate how contextual enrichment helps with emotion recognition across diverse scenarios, comparing the context generated by the large VLM (LLaVA) and the small VLM (SMOLVLM), as well as the predicted labels for each scenario. Additionally, we analyze representative failures from the predictions generated by the model, where CAREFL incorrectly assigned emotion labels, to show how challenging the EMOTIC dataset is. The results are shown in Figure 3.

Qualitative Results

For the first case (Figure 3a), we analyze the indoor multiperson setting. LLaVA generates a detailed narrative capturing different cues in the scene, identifying that it is a social gathering and the presence of laptops, even the model misidentifies the object that the person is holding. The suggested emotions generated help the small model correctly identify all of the labels. In contrast, the SMOLVLM generates a short description, and this model focuses on the person environment, indicating that it is standing near a window, with this description the model also fails to identify the current scenario, indicating that the individual is seeing through the windows, but the suggested emotions guide in a better way, generating 3 of the 5 correct labels.

In the second image (Figure 3b), we have the ambiguous facial expression case, as the person in the image is a man who is standing in what appears to be an art room, contemplating something that is not shown in the image, with no strong facial expressions. In this case, the LLaVa model generates a description emphasizing the solitude and disconnection that the person is feeling, suggested by the environment. Even when the model mentions other emotions like relaxation or enjoyment, this description leads the small model to the correct label predictions. In the alternative scenario, the context generated by SMOLVLM fails to capture the environmental cues, indicating that the person is surrounded by nature, but correctly identifies that the person is having a time alone, leading the small model to predict just one correct label.

In the last case (Figure 3c), the background-driven cues play an important role, because just the facial expressions don't show the correct emotions. SMOLVLM's compact context identifies the task of cooking and the environment correctly, along with the correct labels. In contrast, the context generated by LLaVa doubts about the engagement and esteem emotions, leading to the identification of just the Anticipation and Confidence labels. This highlights that in some scenarios, the detailed descriptions may focus too narrowly, whereas lighter descriptions can generalize and better guide the inference of the small model. This qualitative evaluation confirms that contextual enrichment enhances CAREFL's ability to predict nuanced emotions based on the contexts generated by other models, particularly in ambiguous or low-visibility settings. Importantly, these qualitative findings mirror the quantitative improvements reported in the Ablation Study, where context consistently boosted

F1-scores (up to 50.36% on EMOTIC), demonstrating how context resolves ambiguity and strengthens multi-label emotion prediction.

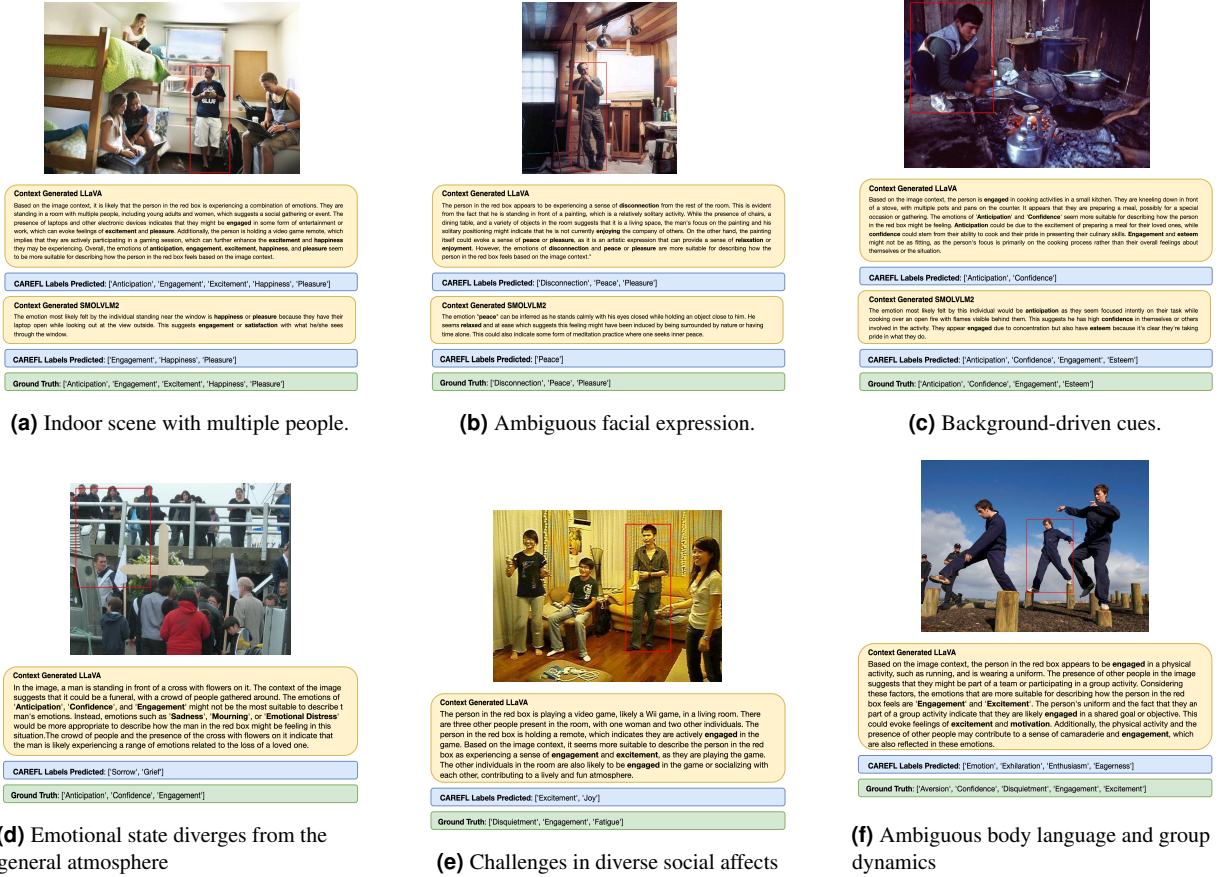


Figure 3. Qualitative examples illustrating CAREFL’s ability to leverage context for accurate emotion recognition in complex scenarios (a–c), and CAREFL’s inference errors (d–f).

Qualitative Error Analysis

In the first example (Figure 3d), we have a person whose emotional state diverges from the general atmosphere, which is a funeral, the person appears to be part of the staff that is helping with the service. In this case, the context generated identifies the correct environment, but suggests that emotions like Anticipation, Confidence, and Engagement are not suitable for the scene and the relation (which are the correct labels), showing that even identifying the correct atmosphere, the emotions could be more complex to identify in terms of analyzing individual persons in scenarios with multiple people.

For the second scenario (Figure 3e), the image shows a group of people playing a video game, in this particular case, the LLaVA model missed cues of social fatigue and discomfort, perhaps because these emotions are less visually salient or contextually hidden by the overall engagement, emphasizing the challenge of distinguishing between diverse social affects.

Finally, in the third image (Figure 3f), we have the scenario of ambiguous body language and group dynamics, the CAREFL framework failed to capture the mix of emotions like Aversion and Disquietment, due to the ambiguous body expressions and the inability to identify the correct relation of the people in the background (security guards) and the people in the front (persons to appears to be prisoners), highlighting the difficulty of modeling subtle interpersonal and negative affect in active scenarios.

Overall, this qualitative error analysis reveals that large VLMs still struggle with fine-grained emotional reasoning at individual and relational levels. In particular, negatively valenced affects remain difficult to detect when they are overshadowed by broader scene cues or ambiguous body language. These results suggest that future improvements for context generation should focus on improving the modeling of interpersonal dynamics, describing the role relationships, and refining the affect interpretation mechanism.

Conclusions

We introduced CAREFL, a framework for emotion recognition that is efficient and aware of context. It combines the expressive reasoning capabilities of large VLMs with lightweight, federated fine-tuning of smaller VLMs. By separating contextual understanding from adaptation, CAREFL allows for privacy-preserving emotion recognition while cutting down on computational, memory, and communication costs.

A key design principle of CAREFL is the division of roles between models. We use high-capacity VLMs, like LLaVA, for offline contextual inference. They generate rich semantic descriptions of scenes without needing training. An SVLM, the only component fine-tuned during training, consumes these contextual signals. This separation avoids the need to adapt large models while still benefiting from their semantic expressiveness. We improve efficiency further through parameter-efficient fine-tuning (LoRA) and 4-bit quantization (QLoRA). Our experiments show this combination reduces GPU memory use from 10.98 GB to 5.96 GB while maintaining competitive metrics and improving inference speed. Compared to full fine-tuning and standard LoRA-based federated training, CAREFL strikes a good balance between efficiency and performance. It shows that quantization does not significantly harm emotion recognition when linked with contextual enrichment.

In addition to efficiency, CAREFL enhances classification performance through semantic contextualization. Ablation studies indicate that contextual input is the most significant factor, boosting the F1-score from 4.74% (no context) to over 50%. Context generated by LLaVA consistently outperforms that from SMOLVLM2. This confirms that richer and more detailed scene-level semantics better support emotion disambiguation, particularly in visually ambiguous situations, as shown in the EMOTIC dataset. LoRA-based adaptation allows the SVLM to specialize effectively in emotion recognition with minimal parameter updates, while federated learning improves generalization. Although federated optimization provides moderate absolute gains, it consistently enhances recall and F1-score. This suggests that exposure to various client data distributions, even with partial participation, acts as a regularizer. It helps reduce overfitting and boosts robustness, especially under non-IID conditions. Our findings extend beyond the EMOTIC dataset, which is complex and relies heavily on contextual reasoning with a multi-label structure.

On CAER-S, a simpler single-label dataset, CAREFL maintains strong performance even with extreme non-IID partitioning ($\alpha = 0.05$), achieving balanced recall and F1-scores. This shows that CAREFL generalizes well across datasets with different label structures, difficulty levels, and data distributions. Across both datasets, the framework remains stable under various federated conditions, from institution-scale deployments with data-rich clients to large-scale decentralized settings with little data per client. This highlights its applicability for real-world scenarios. CAREFL demonstrates that emotion recognition can succeed without centralized training or intensive fine-tuning of foundation models. It offers an alternative for deploying emotion recognition systems in real-world environments like healthcare, education, and personal devices, where data is naturally decentralized and computational resources may be limited. By integrating context generation from frozen VLMs, lightweight SVLM adaptation, quantization optimization, and federated learning, CAREFL provides a practical and scalable solution for privacy-sensitive and resource-constrained situations.

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Author contributions statement

J. L. collected the material and conducted the experiments. All the authors conceived the experiments, analyzed the results, and wrote the first draft of the manuscripts. C. R., A. F., and B. F. reviewed the manuscript. All authors read and approved the final manuscript.

Additional information

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